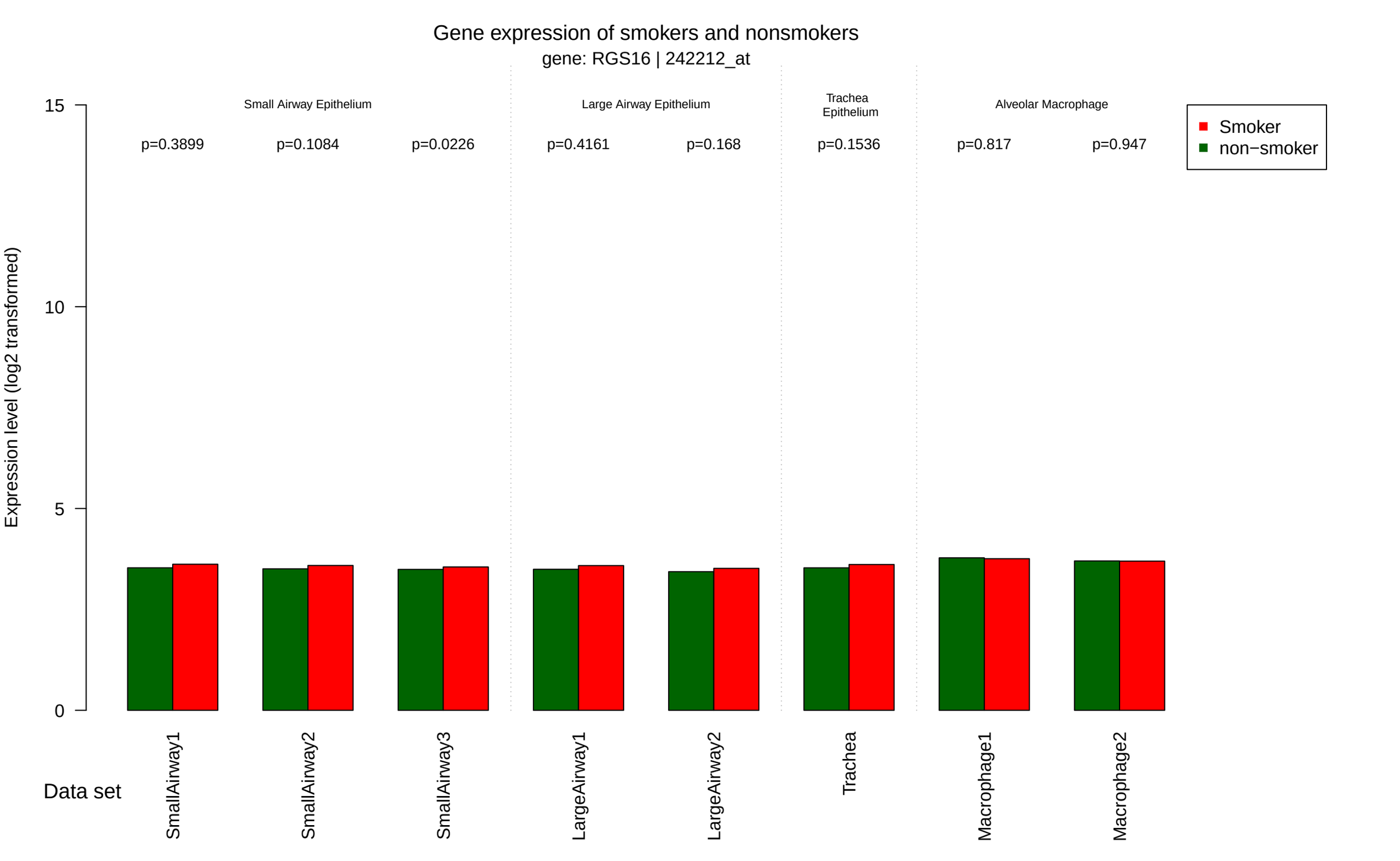
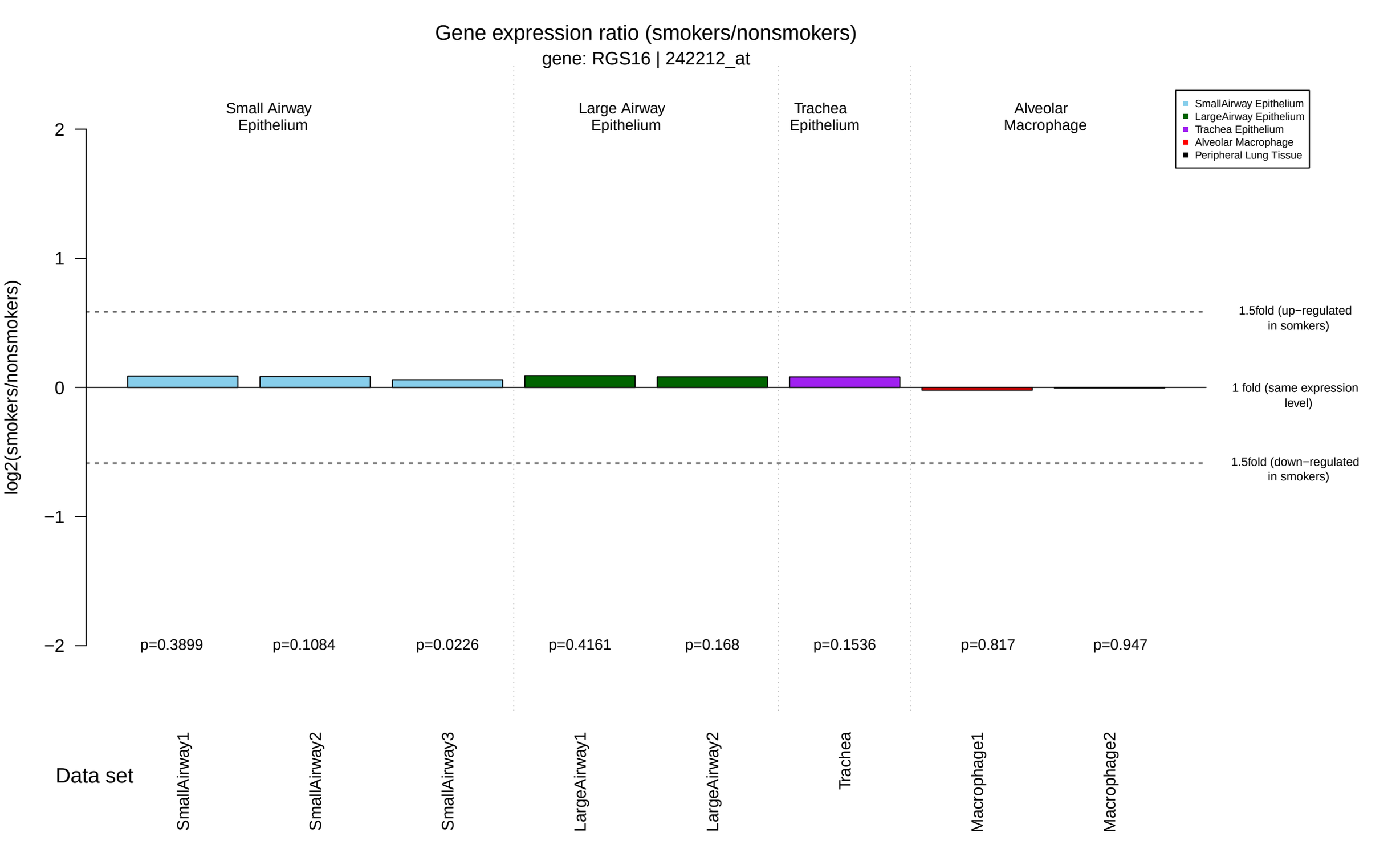
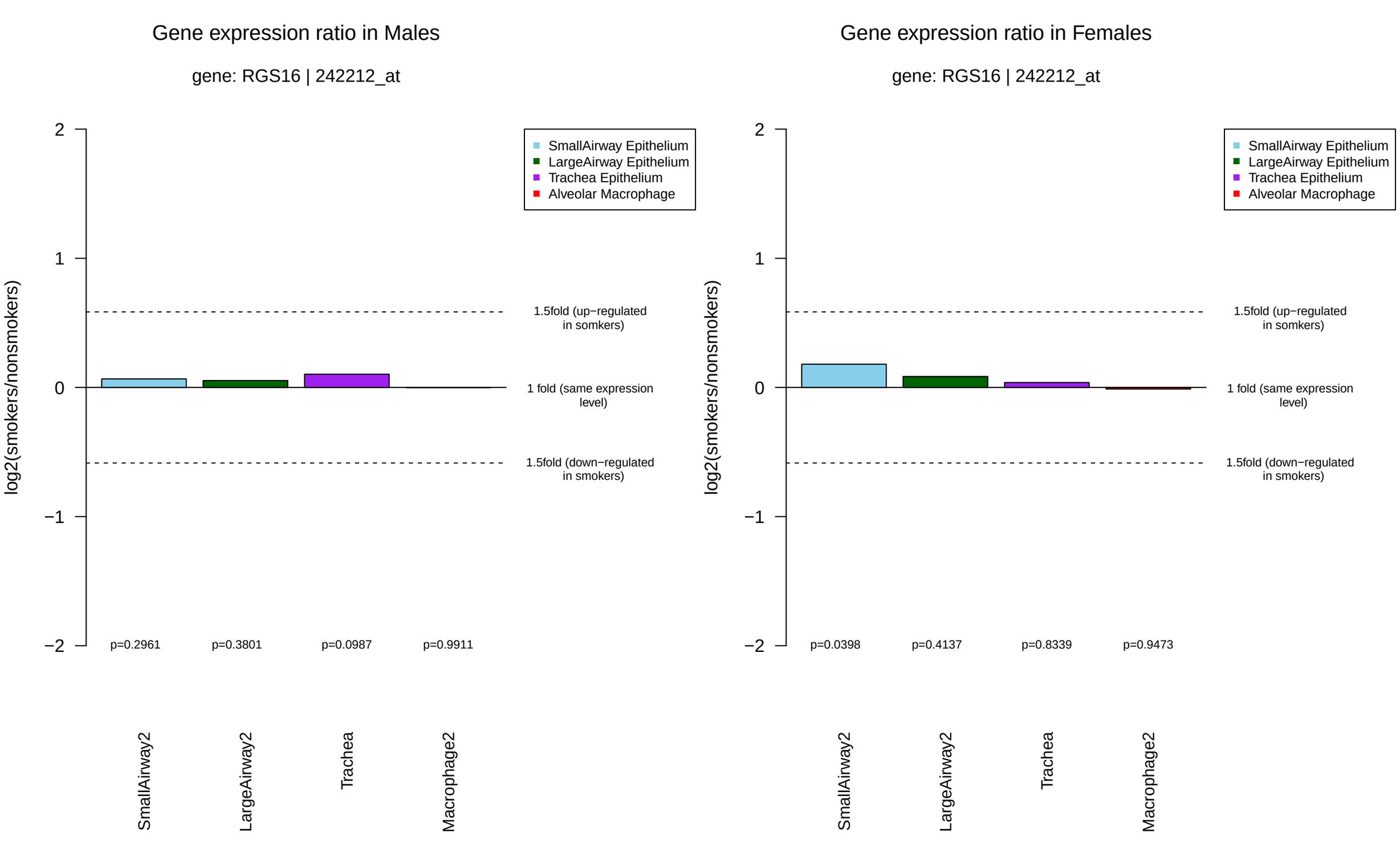
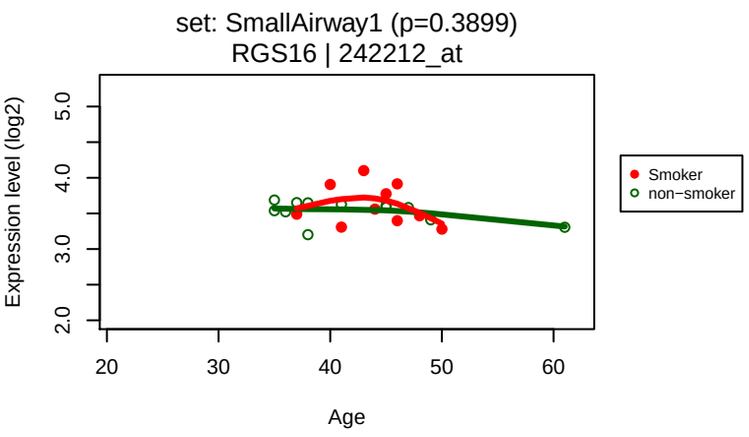
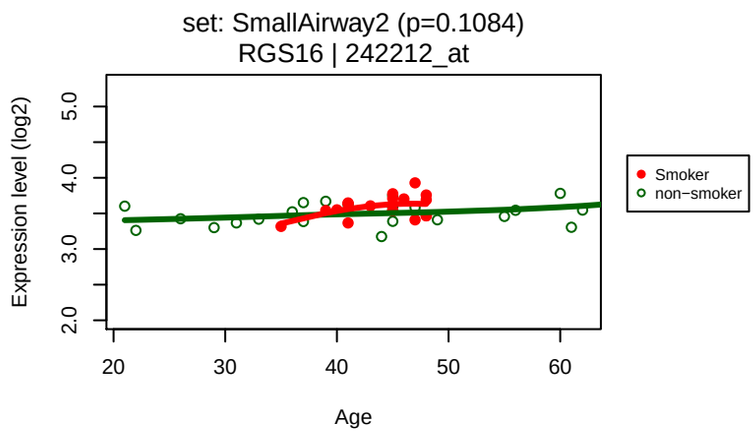
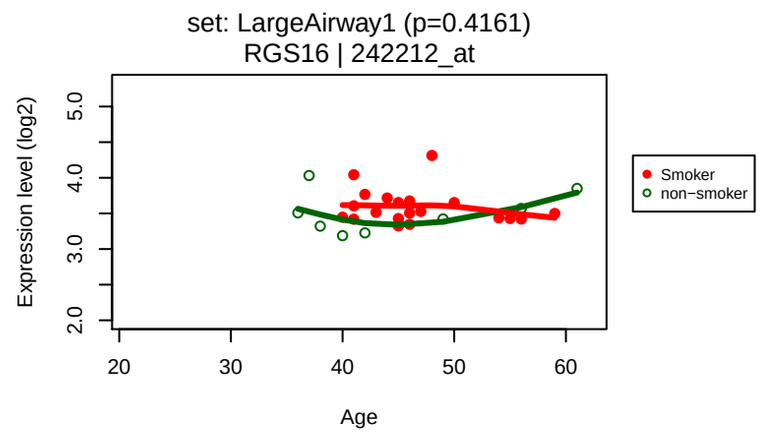

Gene expression of smokers and nonsmokers
gene: RGS16 | 242212_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3899
p=0.1084
p=0.0226
p=0.4161
p=0.168
p=0.1536
p=0.817
p=0.947
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

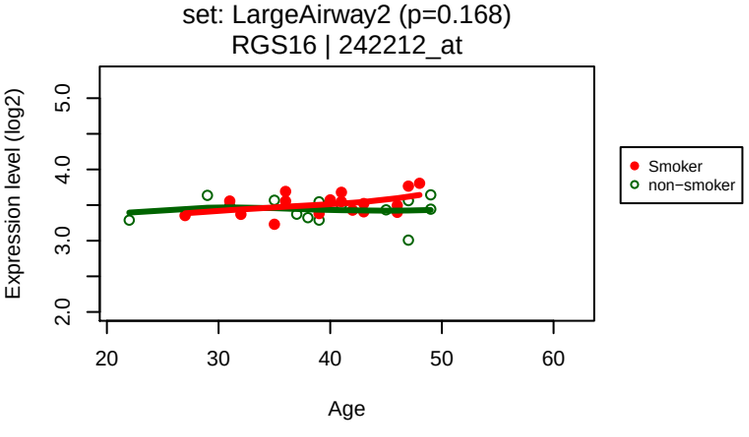

Gene expression ratio (smokers/nonsmokers)
gene: RGS16 | 242212_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3899
p=0.1084
p=0.0226
p=0.4161
p=0.168
p=0.1536
p=0.817
p=0.947
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

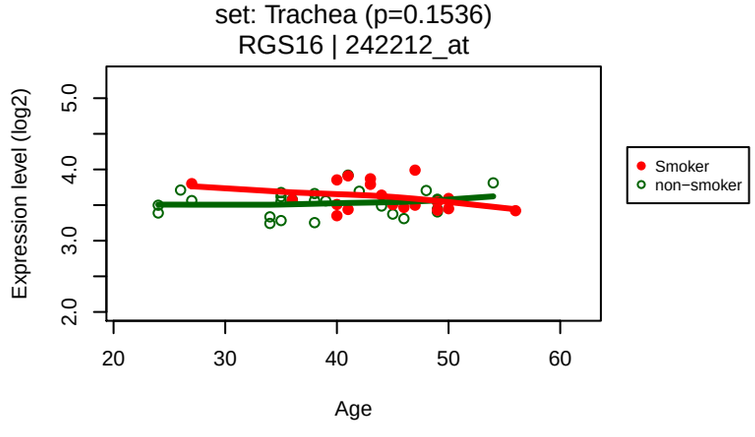

Gene expression ratio in Males
gene: RGS16 | 242212_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2961
p=0.3801
p=0.0987
p=0.9911
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RGS16 | 242212_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0398
p=0.4137
p=0.8339
p=0.9473
SmallAirway2
LargeAirway2
Trachea
Macrophage2

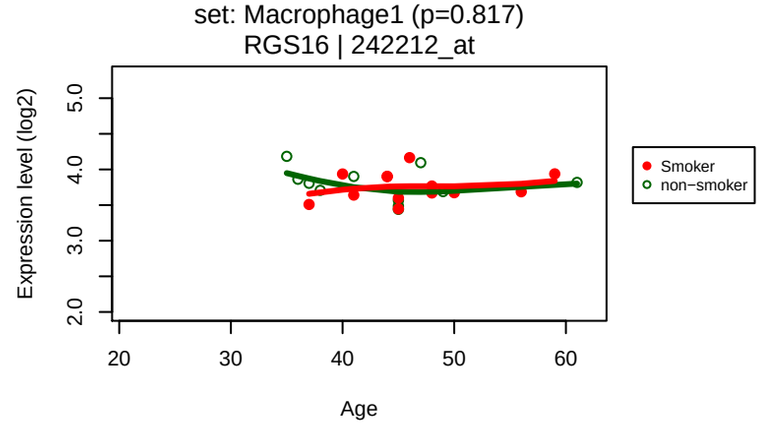

set: SmallAirway1 (p=0.3899)
RGS16 | 242212_at
Expression level (log2)
Age
Smoker
non−smoker

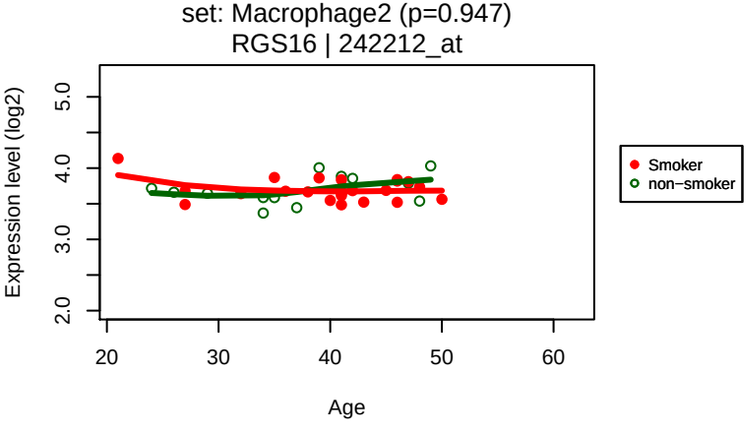

set: SmallAirway2 (p=0.1084)
RGS16 | 242212_at
Expression level (log2)
Age
Smoker
non−smoker

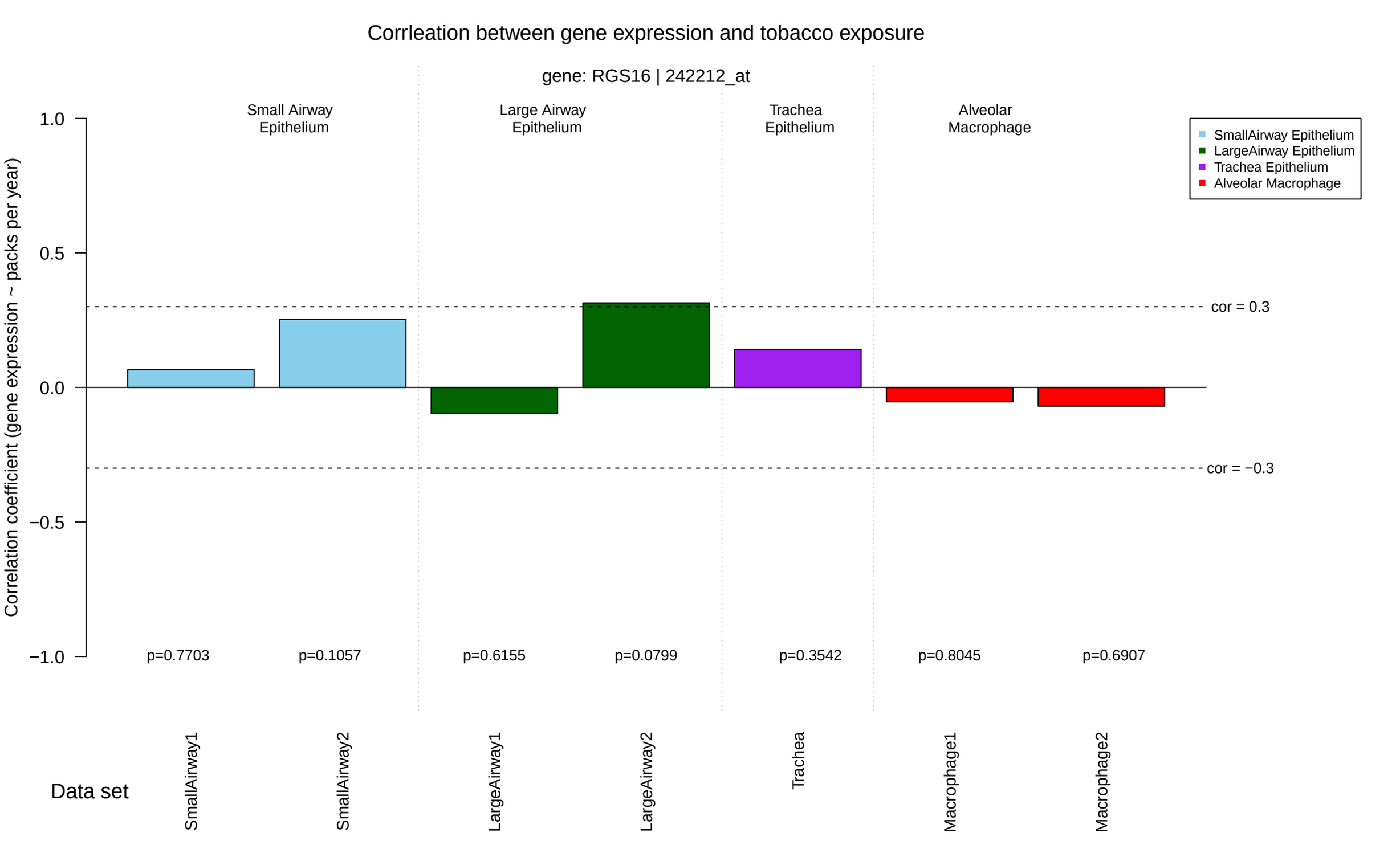

set: LargeAirway1 (p=0.4161)
RGS16 | 242212_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.168)
RGS16 | 242212_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1536)
RGS16 | 242212_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.817)
RGS16 | 242212_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.947)
RGS16 | 242212_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RGS16 | 242212_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7703
p=0.1057
p=0.6155
p=0.0799
p=0.3542
p=0.8045
p=0.6907
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2